



Sudoku Solver Agent

using CSP and Backtracking Search

Artificial Intelligence - CCE414



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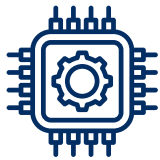
Introduction

- **Approach:** Modeled Sudoku as a **Constraint Satisfaction Problem (CSP)**.
- **Goal:** Fill a **9x9 grid** with numbers **from 1 to 9**.
- **Constraints:** No repeating numbers in the **same Row, Column, or 3x3 Block**.

Algorithm & MRV Optimization



Core Engine:



Recursive Depth-First Search (DFS) combined with **Backtracking**.

Optimization:

Minimum Remaining Values (MRV) Heuristic.

MRV

How it Works:

The agent selects the empty cell with the fewest available options.

If **0 options** exist → **Immediate Backtrack**

If **1 option** exists → **Immediate Assignment..**

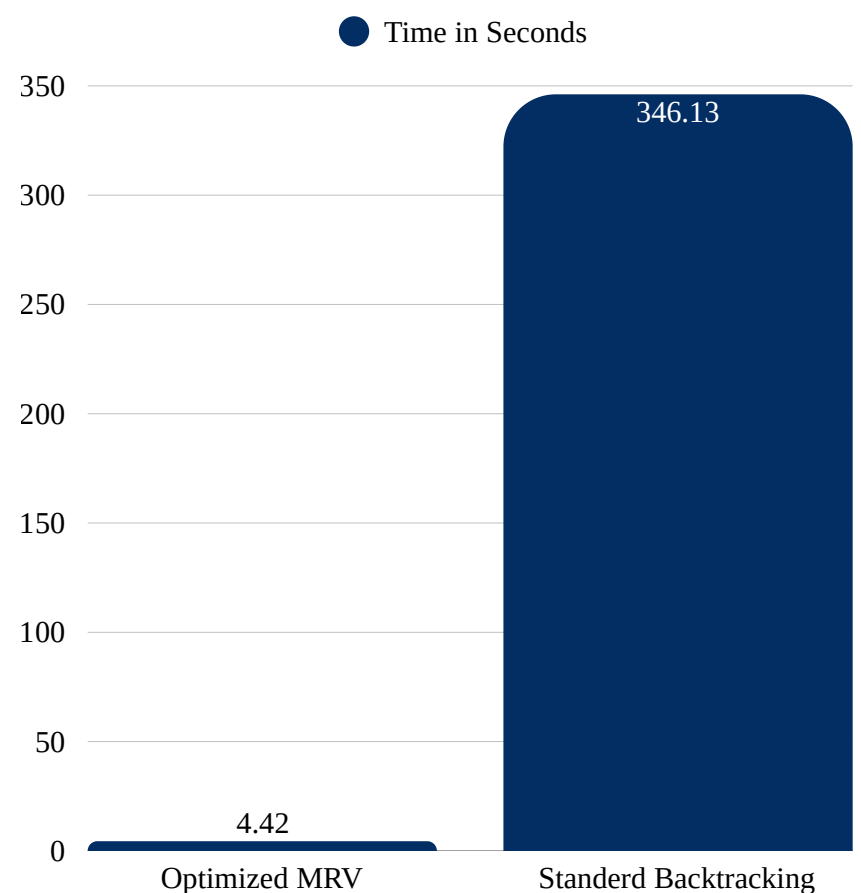
Performance Results

1. **The Test:** A heavily constrained "**Evil Grid**" designed to trap standard algorithms.
2. **Standard Backtracking:** Took **346.13 seconds**.
3. **Optimized MRV Agent:** Took only **4.42 seconds**.
4. **Conclusion:** The MRV heuristic makes the AI agent 78 times faster on complex puzzles.

78x



Try it Yourself



Free Topic:

Concept: A search algorithm that tracks only the best k states in memory, saving massive computing space.

Application: Highly efficient for the **8-Queens Problem**, where rules allow for smooth, step-by-step improvements.



Local Beam Search

PROS:

- Low Memory Usage
- Fast Execution
- Dynamic Allocation

CONS:

- Not Complete
- Not Optimal
- Local Maximum Risk